
Reproducing ALDA zero-shot CV generalization

Arsenii Varaksin^{*1}

Abstract

This work reproduces the results of ALDA, a method for improving zero-shot generalization in reinforcement learning. We compare it to the SAC baseline for a single case of Cartpole environment with OOD evaluation on a color-shifted environment. The results are ambiguous.

1. Introduction

Generalization in computer vision and reinforcement learning is an open problem. Many methods succeed in train environments but fail to generalize to test environments that show data shift. In the CV domain, this can be a change in the background, color, or texture of the objects. ALDA is a method that aims to improve zero-shot generalization in RL by augmenting the training data with adversarial examples. We reproduce ALDA and evaluate it on the Cartpole environment with a color shift in the test environment. The results are ambiguous, and we discuss possible reasons for this.

1.1. SAC

Soft Actor-Critic (SAC) (Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor, Haarnoja T. et. al.) is a popular off-policy reinforcement learning algorithm that optimizes a stochastic policy in an entropy-regularized framework. It has been shown to achieve state-of-the-art performance (at publication) on various continuous control tasks. We use SAC as our baseline for comparison with ALDA.

1.2. Disentanglement

The task of disentangled representation learning is to learn a representation of the data that separates the underlying factors of variation. As such, it allows to reconstruct the assumed underlying factors of variation in the data in such a way the factors are uncorrelated with each other. Then,

a simple change in data distribution would keep most of the factors unchanged, and the model would be able to generalize to the new distribution. Hopfield networks are commonly used for this purpose. Their main idea is to store and retrieve in-distribution patterns so that for a given OOD input, the network can retrieve the closest stored pattern, which is assumed to be in-distribution.

2. ALDA

ALDA (Zero-Shot Generalization of Vision-Based RL Without Data Augmentation, Batra S. et. al.) combines a disentanglement method (QLAE with modifications) and SAC. QLAE maps visual observations to a learnable latent space. The map is a softmax-weighted combination of memories with logits proportional to (minus) the L_1 -distance (metric chosen due to the disentanglement paradigm). Each latent dimension corresponds to a separate set of memories and thus the model is encouraged to learn disentangled dimensions.

A set of memories (e.g. random uniform) is initialized. Then, an encoder learns to map visual observations in such a way that they are close to the QLAE reconstruction.

The QLAE memories, and a supporting reconstruction model, are updated to minimize reconstruction loss.

As in the original paper, gradients are not propagated from critic to the encoder. As such, policy and critic are trained on the convolution of QLAE latent reconstruction by a trainable 1D network.

3. Experiments

We evaluate ALDA and SAC+Autoencoder (SAC+AE) on the Cartpole environment. A single run of each method with the random seed “1” was done. The model is trained on the Cartpole environment and evaluated on the “color-hard” Cartpole environment which randomizes the color of the agent and background to extreme RGB values according to the original paper.

The two methods were trained for about 58k environment steps (until a timeout on Kaggle free tier). In fact, SAC+AE was trained further, for 100k environment steps total.

^{*}Equal contribution ¹HSE, Moscow, Russia. Correspondence to: Arsenii Varaksin <asvaraksin@edu.hse.ru>.

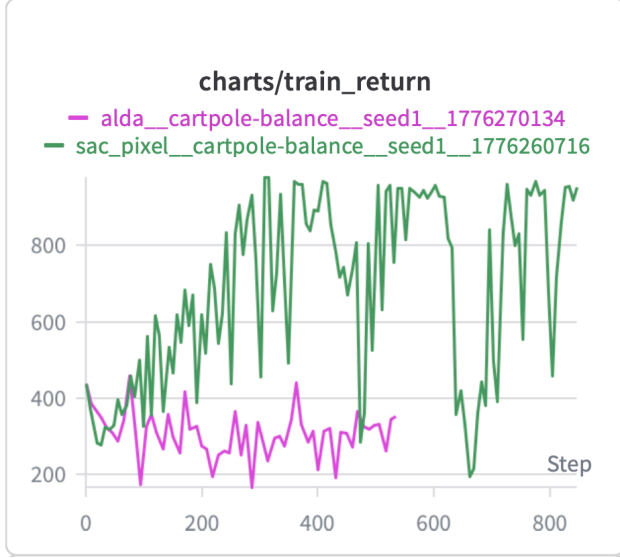


Figure 1. Train returns

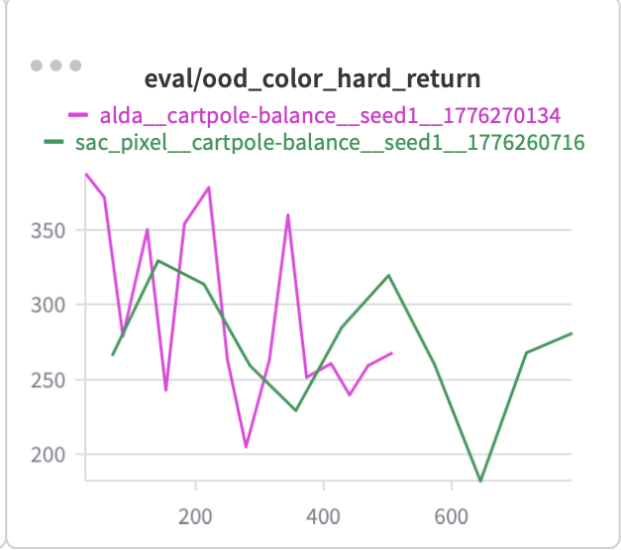


Figure 3. Eval returns

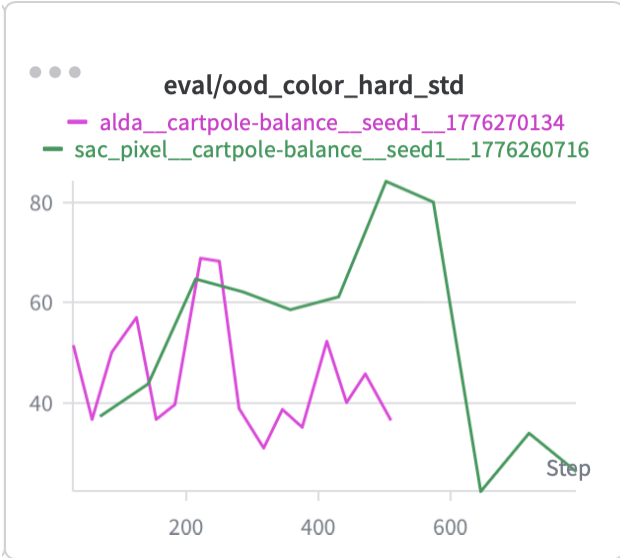


Figure 2. Eval returns std.

Losses, train and eval returns were recorded. In addition, eval returns standard deviation was recorded.

3.1. Results

The results are ambiguous. We were unable to run ALDA to the proposed 500k environment steps due to infrastructure restrictions. The train results of ALDA seemed to be lower than of SAC+AE, probably due to the disentanglement process and limited propagation of gradients from the critic to the encoder.

The eval returns of ALDA were higher than of SAC+AE and the standard deviation was lower. Assuming independence, ALDA beats SAC+AE with $p < 0.32$ and remark-

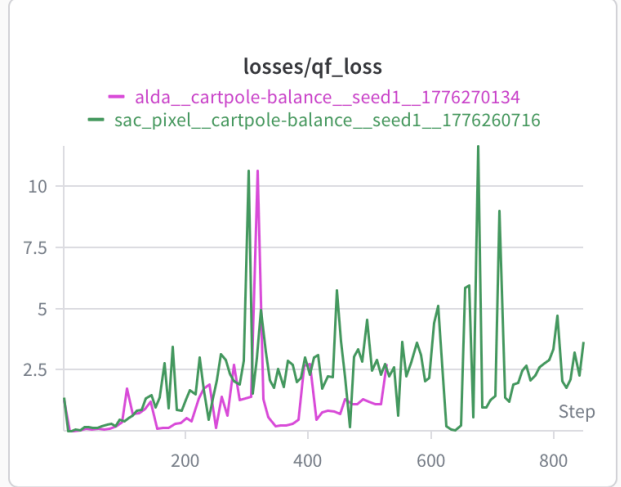


Figure 4. Q loss



Figure 5. Association loss and actor loss

ably has no visible degradation in quality in the OOD setting. However, it seems ALDA policy didn't converge in training time, looking at the policy loss, so a longer training loop could lead to better performance.

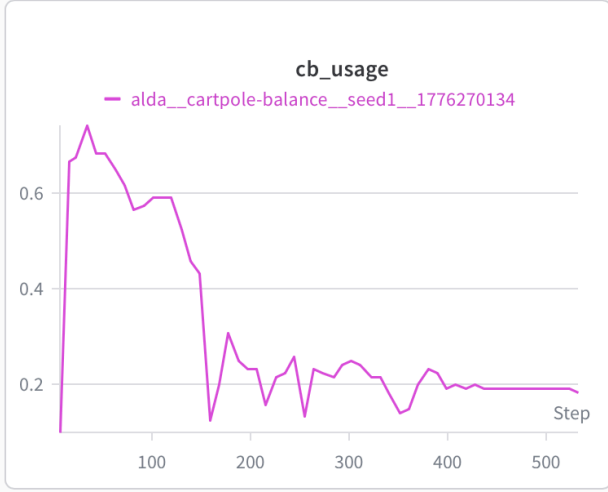


Figure 6. Codebook usage

Codebook usage is estimated as follows. It is considered to be the proportion of memories:

$$\left\{v_{ij} : \exists z_i : j = \operatorname{argmin}_j \|z_i - v_{ij}\|_1\right\} \quad (1)$$

It seems to collapse to a value of 0.2 and possibly lower at the end of the training loop. While a specific target value is not known, we could ask ourselves a question: what does it tell us about the dimensionality of the observations?

A single batch has 768 latent vectors, and each memory set for a latent dimension has 10 memories. Thus, the probability of a memory not to be selected is $(0.9)^{768} < 1e - 10$.

We could hypothesize that the batch size is too large, e.g. latent vectors cluster tightly around N specific values in each batch. Then there would be about $\log_{0.9}(0.8)$ to $\log_{0.9}(0.3)$, $\approx 2.1 - 11.4$ such clusters in each latent dimension. Future analysis might be related to tuning the parameters of the network.

References